The Depolarizing Channel can be Worse than the Dephasing Channel when replacing the Identity map in a Problem of Quantum Control

Aman Chawla

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There are several types of quantum channels mentioned in the text [2]. One way of comparing quantum channels is to look at their maximum input-output fidelities.

Another way, which we employ in this paper, is to use the channel as part of a feedback quantum control system and compute the maximum fidelity achievable.

In this paper we look at a feedback quantum control system described in [1]. The system is as follows:

- A qubit is prepared in a certain state.
- The state is passed through the dephasing channel.
- A weak measurement is applied on the state.
- Based on the measurement result a correction or feedback control is applied on the qubit.

The end-to-end fidelity is then computed and found to be optimal.

We will be interested in making a more realistic model of quantum control where the feedback quantum control (weak measurement followed by control) is itself mediated by a noisy quantum channel. That is, after the weak measurement, the qubit undergoes a CPTP quantum operation before being subjected to the quantum control operation.

In a sense, this is already implemented in [1] because one can view the absence of the CPTP map as the presence of the Identity map. But we will be interested in putting explicit (non-Identity) CPTP maps in between the weak measurement and quantum control steps.

We will study the dephasing map and the depolarizing map, find their optimal fidelities and compare them.

As a first baby step, let us consider inserting first the dephasing and then the depolarizing channel where the Identity map is at present - after the weak measurement and before the quantum control - and compute the fidelity of the scheme in each case. Note that this is *not* the optimal fidelity for each case since one could further optimize the quantum controller specficially for each CPTP map considered. Nevertheless, the calculations are given below, and may be instructive.

1 Dephasing map

The initial state of the qubit is:

$$\begin{aligned} |\psi_1\rangle &= \cos\frac{\theta}{2}(\frac{|0\rangle + |1\rangle}{\sqrt{2}}) + \sin\frac{\theta}{2}(\frac{|0\rangle - |1\rangle}{\sqrt{2}}) \\ &= \alpha|0\rangle + \beta|1\rangle \end{aligned} \tag{1}$$

where

$$\alpha \equiv \frac{1}{\sqrt{2}}(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}) \tag{2}$$

and

$$\beta \equiv \frac{1}{\sqrt{2}}(\cos\frac{\theta}{2} - \sin\frac{\theta}{2}) \tag{3}$$

Thus, the corresponding density operator

$$|\psi_1\rangle\langle\psi_1| = \frac{1}{2}I + \alpha\beta X + \frac{\alpha^2 - \beta^2}{2}Z \tag{4}$$

where we have used the fact that $\alpha^2 + \beta^2 = 1$.

Here we introduce a notation that will help in simpler expressions for density matrices. We rely on the fact that the Pauli matrices form a basis for complex 2×2 matrices and we express a general matrix as a vector in this four-dimensional space, with each coordinate specifying the coefficient multiplying the Pauli matrices I, X, Y and Z in that order.

Thus the density operator above becomes

$$|\psi_1\rangle\langle\psi_1| = \{\frac{1}{2}, \alpha\beta, 0, \frac{\alpha^2 - \beta^2}{2}\} \tag{5}$$

This then goes through the dephasing channel of [1]. The corresponding output is

$$\mathcal{E}(|\psi_1\rangle\langle\psi_1|) = p(Z|\psi_1\rangle\langle\psi_1|Z) + (1-p)|\psi_1\rangle\langle\psi_1| \tag{6}$$

This simplifies to

$$\mathcal{E}(|\psi_1\rangle\langle\psi_1|) = \{\frac{1}{2}, (1-2p)\alpha\beta, 0, \frac{\alpha^2 - \beta^2}{2}\} \tag{7}$$

Next the weak measurement given by measurement operators M_0 and M_1 is applied. We first simplify the expressions for the measurement operator M_0. We obtain,

$$M_0 = \gamma I + \Delta Y \tag{8}$$

where

$$\gamma \equiv (\frac{1}{2}\cos\frac{\chi}{2} + \frac{1}{2}\sin\frac{\chi}{2}) \tag{9}$$

and

$$\Delta = (\frac{1}{2}\cos\frac{\chi}{2} - \frac{1}{2}\sin\frac{\chi}{2}) \tag{10}$$

Thus

$$M_0^\dagger = \gamma I + \Delta Y \tag{11}$$

where we have used the Hermitian-property of the Pauli matrix Y.

The non-normalized (by the probability of its occurrence) post-measurement state is given by

$$\rho_{out}^{(0)} = M_0 \mathcal{E}_p(|\psi\rangle_1 \langle\psi|_1) M_0. \tag{12}$$

This simplifies to

$$\rho_{out}^{(0)} = \{\frac{1}{2}(\gamma^2 + \Delta^2), (1-2p)\alpha\beta(\gamma^2 - \Delta^2), \gamma\Delta, \frac{\alpha^2 - \beta^2}{2}(\gamma^2 - \Delta^2)\} \tag{13}$$

corresponding to the measurement result 0.

This state enters the dephasing feedback channel. The corresponding output is

$$\begin{aligned} \mathcal{E}_{p'}(\rho_{out}^{(0)}) &= (1-p')\rho_{out}^{(0)} + p^{'}(Z\rho_{out}^{(0)}Z) \\ &= \{\frac{1}{2}(\gamma^2+\Delta^2), (1-2p^{'})(1-2p)\alpha\beta(\gamma^2-\Delta^2), (1-2p')\gamma\Delta, \frac{\alpha^2-\beta^2}{2}(\gamma^2-\Delta^2)\} \end{aligned} \tag{14}$$

After this the sub-optimal (because not optimized for this channel) control Z_η is applied.

To compute the corresponding controlled output state, we first compute the effect of this sub-optimal control on the basis states. We find

$$Z_\eta I Z_\eta^\dagger = I \tag{15}$$

$$Z_\eta X Z_\eta^\dagger = X \tag{16}$$

$$Z_\eta Y Z_\eta^\dagger = Y - \sin\eta X \tag{17}$$

and

$$Z_\eta Z Z_\eta^\dagger = Z \tag{18}$$

Thus the controlled output state is easily written down as

$$\rho^{"} = \{\frac{1}{2}(\gamma^2+\Delta^2), (1-2p')(1-2p)\alpha\beta(\gamma^2-\Delta^2)-\sin\eta, -(1-2p')\gamma\Delta, \frac{\alpha^2-\beta^2}{2}(\gamma^2-\Delta^2)\} \tag{19}$$

We now compute the fidelity between this output state and the original input state. It is given by

$$F(|\psi_1\rangle, \rho^{"}) = \langle\psi_1| \rho^{"} |\psi_1\rangle \tag{20}$$

To compute this we first compute the components. Thus

$$\langle\psi_1| I |\psi_1\rangle = 1, \tag{21}$$

$$\langle\psi_1| X |\psi_1\rangle = 2\alpha\beta, \tag{22}$$

$$\langle\psi_1| Y |\psi_1\rangle = 0, \tag{23}$$

and

$$\langle\psi_1| Z |\psi_1\rangle = \alpha^2 - \beta^2. \tag{24}$$

Thus,

$$\langle\psi_1| \rho^{"} |\psi_1\rangle = \frac{1}{2}+2(1-2p')(1-2p)\alpha^2\beta^2(\gamma^2-\Delta^2)-2\alpha\beta\sin\eta+\frac{\alpha^2-\beta^2}{2}(\gamma^2-\Delta^2) \tag{25}$$

where we used the fact that $\gamma^2 + \Delta^2 = 1$.

2 Depolarizing map

The input is

$$\rho_{out}^{(0)} = \{\frac{1}{2}(\gamma^2+\Delta^2), (1-2p)\alpha\beta(\gamma^2-\Delta^2), \gamma\Delta, \frac{\alpha^2-\beta^2}{2}(\gamma^2-\Delta^2)\} \tag{26}$$

which corresponds to measurement result "0".

This enters a depolarizing channel with parameter p". The corresponding output is

$$\begin{aligned}
\mathcal{E}_{p"}(\rho_{out}^{(0)}) &= (1-p")\rho_{out}^{(0)} + \frac{p"}{2}I \qquad (27)\\
&= \{\frac{1}{2}(1-p") + \frac{p"}{2}, (1-p")(1-2p)\alpha\beta(\gamma^2-\Delta^2), (1-p")\gamma\Delta, (1-p")\frac{\alpha^2-\beta^2}{2}(\gamma^2-\Delta^2)\}
\end{aligned}$$

Thereafter the sub-optimal control is performed. The controlled state is found to be

$$\rho^{"} = \{\frac{1-p^{"}}{2}+\frac{p^{"}}{2},(1-p^{"})(1-2p)\alpha\beta(\gamma^2-\Delta^2)-(1-p^{"})\gamma\Delta\sin\eta,(1-p^{"})\gamma\Delta,(1-p^{"})\frac{\alpha^2-\beta^2}{2}(\gamma^2-\Delta^2)\} \tag{28}$$

The fidelity between $\rho^{"}$ and $|\psi_1\rangle\langle\psi_1|$ is found to be

$$F(|\psi\rangle,\rho^{"}) = \frac{1-p^{"}}{2}+\frac{p^{"}}{2}+2(1-p^{"})(1-2p)\alpha^2\beta^2(\gamma^2-\Delta^2)-2\alpha\beta\gamma\Delta\sin\eta(1-p^{"})+(1-p^{"})\frac{(\alpha^2-\beta^2)^2}{2}(\gamma^2-\Delta^2) \tag{29}$$

3 Comparison

The difference between the fidelities in the presence of the dephasing and depolarizing channels is as follows.

$$difference = \frac{p^{"}}{2}+\frac{p^{"}}{2}(\alpha^2-\beta^2)^2(\gamma^2-\Delta^2)+2(p^{"}-2p')(1-2p)\alpha^2\beta^2(\gamma^2-\Delta^2)+2\alpha\beta\sin\eta(\gamma\Delta(1-p^{"})-1) \tag{30}$$

which is clearly non-zero.

Suppose $\gamma^2=\Delta^2$. This happens when $\chi = 0,\pi,2\pi,\ldots,n\pi$. Then,

$$diff = \frac{p^{"}}{2} + 2\alpha\beta\sin\eta(\gamma\Delta(1-p^{"})-1) \tag{31}$$

This is positive when

$$p^{"} > \frac{2\alpha\beta\sin\eta(1-\gamma\Delta)}{\frac{1}{2}-2\alpha\beta\sin\eta\gamma\Delta} \tag{32}$$

Thus the dephasing channel provides greater fidelity as a feedback channel when the above relation holds and the weak measurement is such that $\chi = n\pi$. This is the first comparative result.

References

[1] Agata M. Branczyk, Paulo E. M. F. Mendonca, Alexei Gilchrist, Andrew C. Doherty, and Stephen D. Bartlett. Quantum control of a single qubit. *arxiv:quant-ph/0608037v2*, 2006.

[2] Michael Nielsen and Issac Chuang. *Quantum Computation and Quantum Information.* Cambridge University Press, 2002.